Do World Model Diagnostics Agree? A Cross Diagnostic Validity Study on a Controlled Model Zoo

Venkatesh Nagarjuna, Siddardha Reddy, Nikhil Obuleni, Viharika Vemparala, and Amir Jafari

***Abstract*—Claims that sequence models learn a world model of their domain rest on four families of diagnostics: next token validity, state equivalence, inductive bias probes and representation probing. Each published study uses one of them, on its own models, and on Othello they reach opposite conclusions. We run all four on the same controlled zoo of 81 Othello models (three architectures, three scales, three training distributions, three seeds) under a preregistered analysis, and judge them against a fragility suite frozen before any score. The diagnostics agree strongly on raw scores (Kendall W 0.87), but almost all of this agreement is shared competence: with the legal move rate controlled, W falls to 0.27 and no pair of the diagnostics that can be separated from competence agrees. Their disagreement is not explained by architecture, scale or training data. Only the representation probe adds to the legal move rate in predicting fragility, and an exploratory analysis shows that this signal is a difference between training distributions, not a relation among comparable models. Navigation and a held out program interpreter test whether these findings generalize.**

***Index Terms*—World models, interpretability, probing, measurement validity, Othello, preregistration.**

I. Introduction

WHETHER a sequence model trained only to predict the next token builds a coherent internal model of its domain is a central open question. The same Othello models have been reported both as having a world model [1]–[3] and as lacking a coherent one [4], [5]. Each of these studies uses its own diagnostic on its own models, so their disagreement cannot be interpreted. Following the measurement view of Jacobs and Wallach [6], we treat "has a world model" as a construct and the diagnostics as competing measures of it, and ask whether they agree (convergent validity) and which predicts failure on shifted tasks (predictive validity). We propose no new diagnostic.

Our contributions are: the first head to head comparison of four world model diagnostics on the same models and ground truth; a test of their predictive validity against a fragility suite frozen before any score; and an open benchmark with a common interface to which future diagnostics can be added.

II. Related Work

Representation probing reads the board from activations [1], linearly once squares are labelled relative to the player [2], and across fine tuned language models [3]; state tracking has been tested behaviourally in chess [7]. Vafa et al. derive state equivalence metrics from the Myhill Nerode theorem [4], [8] and propose inductive bias probes [5]. Control tasks [9] and the limits of probing [10] shape our D4. Benchmarks often claim more than they measure [11]. For inference with repeated runs we follow the cluster bootstrap [12] and the case for accounting for seed variance [13].

III. Method

A. Model zoo and domain

Othello's true state (board and player to move) is replayed exactly by our rules engine. We train transformers [14], stacked LSTMs [15] and Mamba [16] at three scales (0.48 to 25.3 million parameters) on synthetic, championship (WTHOR) or mixed games, three seeds each, with one fixed budget. A model is admitted if its validation legal move rate, its *competence*, is at least 0.50; all 81 are admitted (competence 0.559 to 0.969).

B. Diagnostics

D1 is the share of positions whose top predicted move is legal. **D2** is one minus the mean of the compression and distinction errors of Vafa et al. [4] with next move continuations. **D3** is the balanced accuracy gain of the pretrained model over a random weight twin when adapted on 64 examples to "is square X mine?" [5], with a symbolic regression arm [17]. **D4** is the selectivity of a linear mine, theirs or empty probe [9] at the best layer, with a causal edit check [1]. Protocols were tuned only on the two published Othello GPTs.

C. Fragility and analysis

The fragility suite keeps the rules and shifts the situations (other distributions, after a pass, symmetries, forced detours); relative fragility divides the mean drop in legal move rate by the model's own rate. All rules were preregistered with git tags before the results they affect. H1 uses Kendall's tau [18], Kendall's partial tau given competence [19] and Kendall's W [20]; H2 regresses per model disagreement on the design factors against a seed noise floor; H3 tests D2 to D4 beyond D1 by partial Spearman correlation. Intervals are 95% cluster bootstrap intervals over the 27 configurations.

The authors are with the Data Science Program, The George Washington University, Washington, DC, USA.

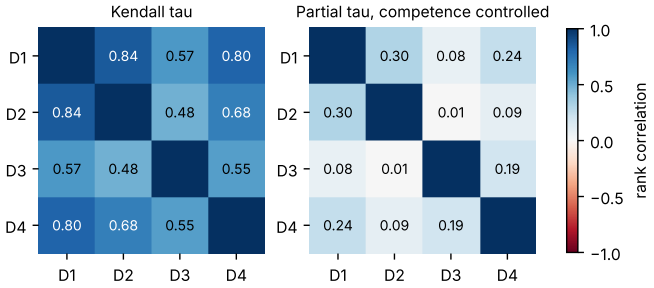


Fig. 1. Rank agreement between diagnostics, raw and with competence controlled (81 models).

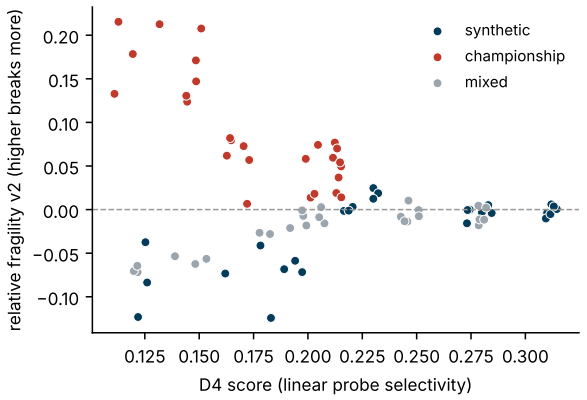


Fig. 2. D4 against relative fragility, by training distribution (exploratory).

IV. Results on Othello

H1. Five of six pairs agree on raw scores and W is 0.872 [0.764, 0.921] (Fig. 1). Every diagnostic follows competence (tau 0.968 for D1, 0.826 D2, 0.790 D4, 0.577 D3). D1 pairs are not interpretable after the control (preregistered rule), and the three pairs among D2, D3 and D4 all lose their agreement; W on competence residuals is 0.267 [0.189, 0.427].

H2. Mean disagreement is 0.089 against a seed noise floor of 0.028; no factor is significant after Holm correction (architecture $p = 0.78$, scale $p = 0.081$, training data $p = 0.34$).

H3. D1 does not predict relative fragility (0.150 [−0.315, 0.588]). Only D4 adds to D1 (partial −0.554 [−0.734, −0.124]), but no single diagnostic wins. In an exploratory analysis, championship trained models are both the most fragile and the least probe readable (Fig. 2); within each training distribution the D4 link vanishes, and with training data controlled it is +0.231 [−0.336, 0.589].

V. Discussion

On Othello the diagnostics agree mainly because better models score higher on all of them; beyond competence they measure different things, and none reliably predicts fragility among comparable models. A board that can be read out is not a board that is used: every model is decodable but only 10 of 81 pass the causal check. Studies that report a single diagnostic on models of different quality mostly report competence.

VI. Conclusion and Future Work

We will repeat the analysis on navigation, reproducing the state equivalence results of Vafa et al. [4], and test generalization (H4) on a held out program interpreter.

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